Vector Calculus and Complex Analysis Homework 10

1. By evaluating the contour integral $\int_{|z|=1}\left(z+\frac{1}{z}\right)^{2n}\frac{dz}{z}$ where $|z|=1$ in counterclockwise direction, show that $\int_0^{2\pi}\cos^{2n}\theta d\theta = 2\pi\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$

Solution

$$\int_{|z|=1}\left(z+\frac{1}{z}\right)^{2n}\frac{dz}{z}=\int_0^{2\pi}(2\cos\theta)^{2n}\frac{ie^{i\theta}d\theta}{e^{i\theta}}=2^{2n}i\int_0^{2\pi}\cos^{2n}\theta d\theta$$

$$\int_{|z|=1}\left(z^{2n}+\cdots+\binom{2n}{n}+\cdots+\frac{1}{z^{2n}}\right)\frac{dz}{z}=\binom{2n}{n}\int_{|z|=1}\frac{dz}{z}=2\pi i\frac{(2n)!}{n!n!}$$

$$2^{2n}i\int_0^{2\pi}\cos^{2n}\theta d\theta=2\pi i\frac{(2n)!}{n!n!}$$

$$\int_0^{2\pi}\cos^{2n}\theta d\theta=2\pi\frac{(2n)!}{n!(2\cdot 4\cdots(2n))}\frac{1}{2^n}=2\pi\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$$

2. Let $P(z)=a_nz^n+a_{n-1}z^{n-1}+\cdots+a_1z+a_0,\ a_n\neq 0,$ such that all of its zeros $c_1,c_2,\cdots,c_n$ lie inside the circle $C: z(t)=Re^{it},\quad 0\le t\le 2\pi.$ Show that

(a) $\frac{P'(z)}{P(z)}=\frac{1}{z-c_1}+\cdots+\frac{1}{z-c_n}$

(b) $\int_C\frac{P'(z)}{P(z)}dz=2n\pi i$ and $\int_C\frac{zP'(z)}{P(z)}dz=-2\pi i\frac{a_{n-1}}{a_n}$

Solution

(a) $P(z)=a_n(z-c_1)(z-c_2)\cdots(z-c_n)$

$$P'(z)=a_n\left[(z-c_2)\cdots(z-c_n)+(z-c_1)(z-c_3)\cdots(z-c_n)+\cdots+(z-c_1)\cdots(z-c_{n-1})\right]$$

$$\frac{P'(z)}{P(z)}=\frac{(z-c_2)\cdots(z-c_n)+(z-c_1)(z-c_3)\cdots(z-c_n)+\cdots+(z-c_1)\cdots(z-c_{n-1})}{(z-c_1)(z-c_2)\cdots(z-c_n)}$$

$$=\frac{1}{z-c_1}+\cdots+\frac{1}{z-c_n}$$

(b) $\int_C \frac{P'(z)}{P(z)}dz = \int_C \frac{1}{z-c_1} + \cdots + \frac{1}{z-c_n}dz = 2n\pi i$

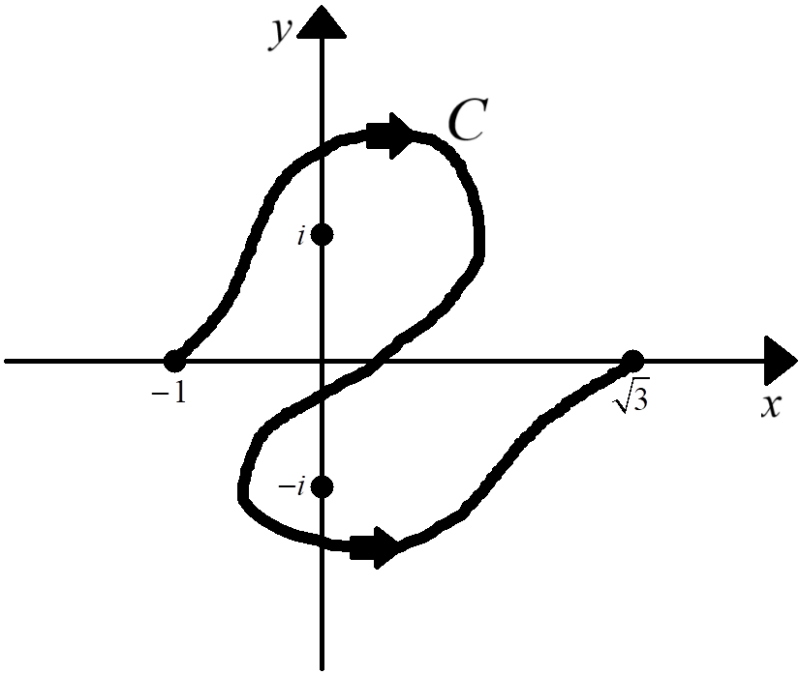
$$\int_C \frac{zP'(z)}{P(z)}dz = \int_C \frac{z}{z-c_1} + \cdots + \frac{z}{z-c_n}dz$$

$$= \int_C n + \frac{c_1}{z-c_1} + \cdots + \frac{c_n}{z-c_n}dz$$

$$= 2\pi i\left(c_1 + c_2 + \cdots + c_n\right)$$

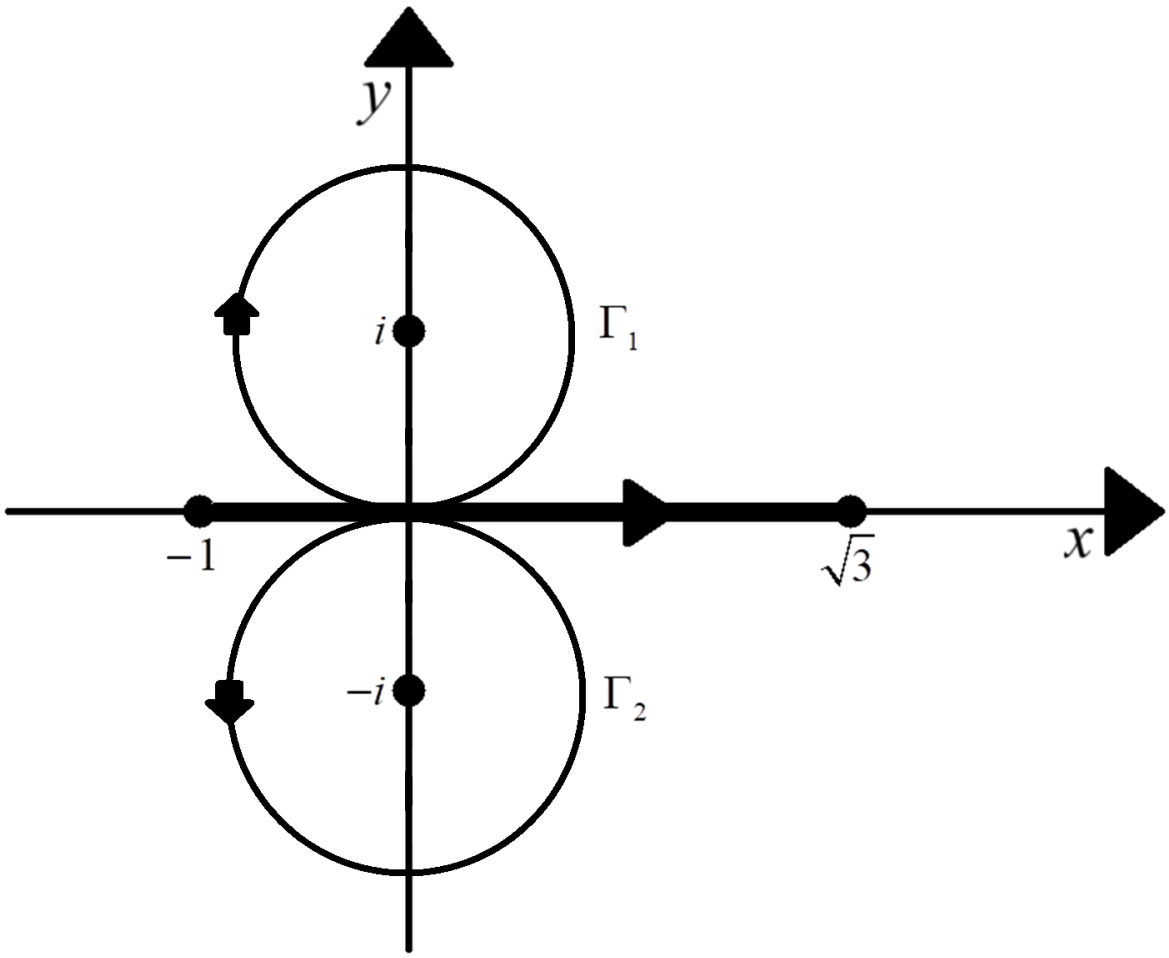
$$= -2\pi i \frac{a_{n-1}}{a_n}$$

3. Calculate $\int_C \frac{1}{z^2+1}dz$ where C is the contour in the following figure.



Solution

The contour C can be continuously deformed to the following picture.



$$\frac{1}{z^2+1}=\frac{A}{z+i}+\frac{B}{z-i}=\frac{A(z+i)+B(z-i)}{z^2+1}$$

$$1=A(z+i)+B(z-i)$$

$$1=2Ai=-2Bi$$

$$\frac{1}{z^2+1}=\frac{i}{2(z+i)}-\frac{i}{2(z-i)}$$

$$\begin{aligned}\int_C \frac{1}{z^2+1}dz &= \int_{\Gamma_1}\frac{1}{z^2+1}dz+\int_{\Gamma_2}\frac{1}{z^2+1}dz+\int_{-1}^{\sqrt{3}}\frac{1}{t^2+1}dt\\ &= \int_{\Gamma_1}-\frac{i}{2(z-i)}dz+\int_{\Gamma_2}\frac{i}{2(z+i)}dz+\int_{-1}^{\sqrt{3}}\frac{1}{t^2+1}dt\\ &= -\frac{i}{2}(-2\pi i)+\frac{i}{2}2\pi i+\tan^{-1}t\Big|_{-1}^{\sqrt{3}}\\ &= -2\pi+\frac{\pi}{3}-\left(-\frac{\pi}{4}\right)\\ &= -\frac{17\pi}{12}\end{aligned}$$

4. Calculate $\int_C \frac{ze^{z^2}}{z^2-1}dz$ where $C=\{(2\cos t, 2\sin t): 0\le t\le \pi\}$.

Solution

$z(t)=2\cos t+2i\sin t=2e^{it}, \quad 0\le t\le\pi, \quad w(t)=z(t)^2=4e^{2it}, \quad 0\le t\le\pi$

That means, $z(t)$ is an upper semi-circle of radius 2 then $w(t)$ is a circle of radius 4.

$$\int_C \frac{ze^{z^2}}{z^2-1}dz=\frac{1}{2}\int_{|w|=4}\frac{e^w}{w-1}dw=\frac{1}{2}2e\pi i=e\pi i$$

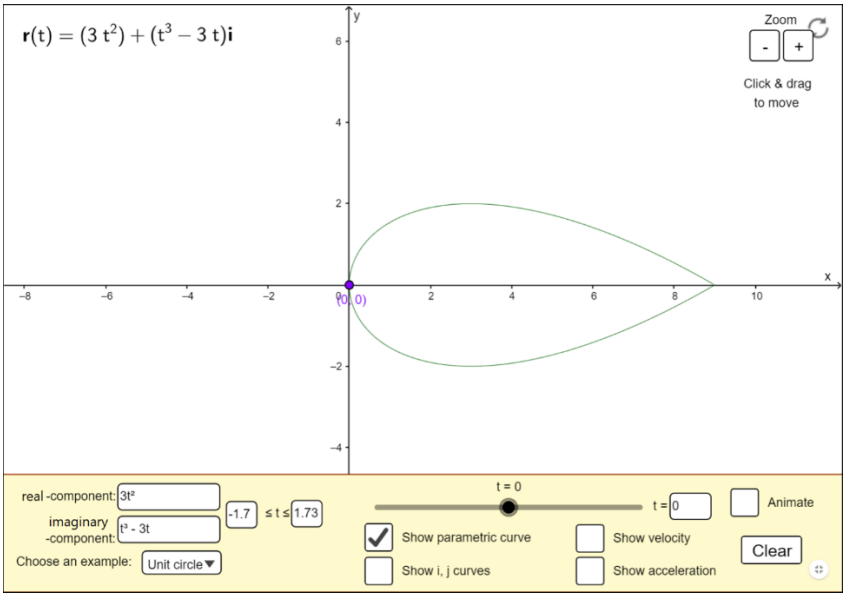
5. Without solving the following contour integrals, show that

(a) $\left|\int_0^{1+i} z^2dz\right|\le 2\sqrt{2},$

(b) $\left|\int_{|z-1|=1}\frac{e^z}{z+1}dz\right|\le 2\pi e^2,$ in counterclockwise direction

(c) $\left|\int_{|z|=R}\frac{z^n}{z^m-1}dz\right|<_{\text{strict}}\frac{2\pi R^{n+1}}{R^m-1}, \quad R>1, m\ge 1, n\ge 0,$ in counterclockwise direction

(d) $\left|\int_C \frac{Arg(z-i)}{z-1}dz\right|\le 6\sqrt{3}\pi$ where $C: z(t)=3t^2+(t^3-3t)i, -\sqrt{3}\le t\le\sqrt{3}.$



Solution

(a) $\left|\int_0^{1+i} z^2 dz\right| \le |1+i|^2 \, \ell\left(\overline{0,1+i}\right) = 2\sqrt{2},$

(b) $$|e^z| = \left|e^{\operatorname{Re} z + i \operatorname{Im} z}\right| = \left|e^{\operatorname{Re} z} e^{i \operatorname{Im} z}\right| = \left|e^{\operatorname{Re} z}\right| \le e^2,$$

since $\operatorname{Re} z \le 2$ on the circle.
And for the denominator in the integrad we have

$$|1+z| = |2 + \cos t + i \sin t| \ge \operatorname{Re}(2 + \cos t + i \sin t) = 2 + \cos t \ge 1,$$

Thus

$$M = \left|\frac{e^z}{1+z}\right| = \frac{|e^z|}{|z+1|} \le \frac{e^2}{1} = e^2.$$

Becauce the integrationcontour is a circle with radius 1 we have that $L = 2\pi$.
$ML-$ estimate gives

$$\left|\oint_{|z-1|=1} \frac{e^z}{z+1} dz\right| \le 2\pi e^2.$$

(c) If $|f(z)| \leqslant m_0 < m$ on an arc γ_0 of the circle, $\gamma = \gamma_0 \cup \gamma_1$, then $\left|\int_\gamma f(z)\, dz\right| = \left|\int_{\gamma_0} f(z)\, dz + \int_{\gamma_1} f(z)\, dz\right| \leqslant m_0 \text{length}(\gamma_0) + m \text{length}(\gamma_1) = (m_0 - m)\,\text{length}(\gamma_0) + m\text{length}(\gamma) < m\text{length}(\gamma)$
Thus if $\left|\int_\gamma f(z)\, dz\right| = \text{ML}$, then $|f(z)| = M$ on γ. In this case, $|z^m| = R^n$, $\left|\frac{1}{z^m - 1}\right| \leqslant \frac{1}{R^m - 1}$, with strict inequality unless $z = \text{R} \cdot \underbrace{e^{2\pi i k/m}}_{m^{th} \text{ root of } w}$. $\therefore M = \frac{R^n}{R^m - 1}$, $L = 2\pi R$, get $\left|\int_{|z|=R} \frac{z^n}{z^m - 1} dz\right| < \frac{R^n}{R^m - 1} \cdot 2\pi R = \frac{2\pi R^{n+1}}{R^m - 1}$, $R > 1$, $m \geqslant 1$, $n \geqslant 0$.

(d) $i(t-i)^3 = i(t^3 - 3it^2 + 3i^2t - i^3) = i(t^3 - 3it^2 - 3t + i) = 3t^2 - 1 + (t^3 - 3t)i = z(t) - 1$

$|z(t) - 1| = |t - i|^3 \geq |i|^3 = 1$

$$\left|Arg(z-i)\right| \leq \frac{\pi}{2}$$

$$\left|\frac{Arg(z-i)}{z-1}\right| \leq \frac{\pi}{2} = M$$

$$L = \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{x'(t)^2 + y'(t)^2}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{(6t)^2 + (3 - 3t^2)^2}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{36t^2 + 9 - 18t^2 + 9t^4}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{9 + 18t^2 + 9t^4}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} (3 + 3t^2) dt$$

$$= \left[3t + t^3\right]_{-\sqrt{3}}^{\sqrt{3}}$$

$$= 12\sqrt{3}$$

$$\frac{\pi^2}{2} = \left|\int_C \frac{Arg(z-i)}{z-1}\, dz\right| \leq ML = \frac{\pi}{2} 12\sqrt{3} = 6\sqrt{3}\pi$$